Homework 1

Module 1 · Homework · No date

Name _____ Date _____

1. A pharmacist dissolved 200 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

Time (hr)	Drug (mcg/mL)
0.5	233.3
1	217.6
2	189.5
4	143.6
8	82.5
12	47.4
16	27.2

1a Module 1

Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Neither: it needs a third plot

B. Zero order

C. First order

1b Module 1

What is the rate constant k for the decomposition of this antibiotic?

1c Module 1

What is the half-life $t_{1/2}$?

1d Module 1

What was the initial volume of water used to prepare the original solution?

1e Module 1

How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

Time (hr)	Drug A (mg)
0.5	489
1	478
2	455
3	433
6	365
9	298
12	230

Time (hr)	Drug A (mg)
16	140

2a Module 1

Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. First order

B. Zero order

C. Cannot tell without logarithms

2b Module 1

What is the rate constant k?

2c Module 1

What is the half-life t½?

3. A solution of a drug was freshly prepared at a concentration of 315 mg/mL. After 30 days at 25°C, the drug concentration in the solution was 75 mg/mL.

3a Module 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

3b Module 1

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: C. First order

Over each 4-hour step from 4 to 16 hr the concentration falls by the same fraction (1.74, 1.74, 1.74), so ln C against time is a straight line. A constant fraction lost per equal time is first order, because the rate is proportional to the amount remaining.

Her homework with her own numbers; her key gives this answer.

1b

Worked by the drill

Answer: 0.1386 hr^{-1}

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: k comes from the slope, not the amount*
temperature	4°C	*not needed*
C_1	217.6 mcg/mL at 1 hr	$\ln(C_1)$, the first point
C_2	27.2 mcg/mL at 16 hr	$\ln(C_2)$, the second point
$t_2 - t_1$	16 hr − 1 hr	the denominator, 15 hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(C_1) - \ln(C_2)}{t_2 - t_1}, \text{ with } t_1 = 1 \text{ hr and } t_2 = 16 \text{ hr}$$

For first order, $\ln C = \ln C_0 - kt$, so ln C against t is a straight line with slope −k, where k is the first-order rate constant. Two points far apart give the slope with the least error; her key uses the 1 hr and 16 hr points, 15 hours apart.

ALGEBRA

$$k = \frac{\ln(217.6) - \ln(27.2)}{15 \text{ hr}} = \frac{2.0794}{15 \text{ hr}} = 0.1386 \text{ hr}^{-1}$$

$\ln(217.6) - \ln(27.2) = \ln(\frac{217.6}{27.2}) = 2.0794$: the mcg/mL cancel inside the logarithm, leaving a pure number. Dividing by the 15 hr between the points leaves reciprocal hours, so k = 0.1386 hr^{-1}, the fraction of the drug lost per hour.

> **DOES THE NUMBER MAKE SENSE?**
>
> From 1 hr (217.6) to 16 hr (27.2) the level falls eightfold, three halvings in 15 hours, so the half-life is near 5 hr and k near $\frac{0.693}{5}$; the two-point slope gives 0.1386 hr^{-1}.

Her homework with her own numbers; her key gives this answer.

1c Worked by the drill

Answer: 5 hr

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: a first-order half-life does not depend on the amount*
table	217.6 mcg/mL at 1 hr and 27.2 mcg/mL at 16 hr	gave k = 0.1386 hr^{-1} in part b, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{k} = \frac{0.693}{0.1386\ hr^{-1}} = 5.00\ hr$$

The half-life t½ is the time for C to fall to half, so $e^{-k\,t½} = 0.5$ and $k\,t½ = \ln 2 = 0.693$, giving $t½ = \frac{0.693}{k}$. With k = 0.1386 hr^{-1} from part b, the reciprocal hours in the denominator invert to hours: 5.00 hr for every halving.

> **DOES THE NUMBER MAKE SENSE?**
> From 1 hr (217.6) to 16 hr (27.2) the level falls eightfold, three halvings in 15 hours, so the half-life is near 5 hr; $\frac{0.693}{0.1386} = 5.00$ hr agrees.

Her homework with her own numbers; her key gives this answer.

1d Worked by the drill

Answer: 800 mL

WHAT THE STEM GIVES

D_0	200 milligrams	written 200,000 mcg, the numerator
C_1	217.6 mcg/mL at 1 hr	brought back 1 hr to C_0 = 250.0 mcg/mL, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$\ln C_0 = \ln C + kt = \ln(217.6) + (0.1386)(1) \rightarrow C_0 = 250.0\ mcg/mL$$

The first-order line $\ln C = \ln C_0 - kt$ rearranges to $\ln C_0 = \ln C + kt$, with C_0 the concentration the moment the drug dissolved. The 1 hr point, 217.6 mcg/mL, has declined for 1 hour at k = 0.1386 hr^{-1}; adding kt = 0.1386 undoes that decline: C_0 = 250.0 mcg/mL.

UNIT

$$200\ mg \times 1000\ mcg/mg = 200{,}000\ mcg$$

The 200 mg dissolved is the amount at time zero. The concentration is in mcg/mL, so the amount is put in micrograms, 200 × 1000 mcg/mg = 200,000 mcg; then mcg over mcg/mL cancels the micrograms and leaves millilitres, the unit asked for the volume.

ALGEBRA

$$V = \frac{200{,}000 \text{ mcg}}{250.0 \text{ mcg/mL}} = 800 \text{ mL}$$

Concentration is amount over volume, $C_0 = \frac{D_0}{V}$, so $V = \frac{D_0}{C_0}$: the whole 200,000 mcg spread through the water at 250.0 mcg/mL. The micrograms cancel and millilitres are left, 800 mL, the volume of purified water the pharmacist used.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 must be a little above the 0.5 hr value of 233.3 mcg/mL; it is 250.0. Then 200,000 mcg at 250.0 mcg/mL is $\frac{200{,}000}{250.0} = 800$ mL.

Her homework with her own numbers; her key gives this answer.

1e

Worked by the drill

Answer: 10 hr

WHAT THE STEM GIVES

D_0	200 milligrams	*not needed: the time to a fraction does not depend on the amount*
fraction	decrease by 75%	leaves 25%, so C_0 over C is 4, two half-lives
table	217.6 mcg/mL at 1 hr and 27.2 mcg/mL at 16 hr	gave $k = 0.1386$ hr^{-1} in part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$t = \frac{\ln(C_0/C)}{k} = \frac{\ln(1/0.25)}{0.1386 \text{ hr}^{-1}}$$

The first-order line $\ln C = \ln C_0 - kt$ solved for the time is $t = \frac{\ln(C_0/C)}{k}$. A 75% decrease leaves 25% of the starting concentration, so C_0 over C $= \frac{1}{0.25} = 4$ whatever the starting value, and $k = 0.1386$ hr^{-1} is the rate constant from part b.

ALGEBRA

$$t = \frac{1.3863}{0.1386 \text{ hr}^{-1}} = 10.00 \text{ hr} = 2 \times 4.999 \text{ hr}$$

$\ln 4 = 1.3863$, a pure number, so dividing by k in hr^{-1} leaves hours: $\frac{1.3863}{0.1386 \text{ hr}^{-1}} = 10.00$ hr. Falling to 25% is two halvings, 100% to 50% to 25%, so the time is exactly $2 \times t_{½} = 2 \times 4.999$ hr, which checks the answer against part c.

> **DOES THE NUMBER MAKE SENSE?**

Losing 75% leaves a quarter, two halvings, so the time is two half-lives: 2 × 4.999 hr = 10.00 hr, more than the one half-life of 4.999 hr.

Her homework with her own numbers; her key gives this answer.

2a

Worked by the drill

Answer: B. Zero order

Each 3-hour step from 3 to 12 hr loses about the same 67–68 mg, so the amount against time is a straight line on ordinary axes. A constant amount lost per equal time is zero order, because the rate does not depend on how much drug is left.

Her homework with her own numbers; her key gives this answer.

2b

Worked by the drill

Answer: 22.5 mg/hr

WHAT THE STEM GIVES

A1	478 mg at 1 hr	the first amount
A2	230 mg at 12 hr	the second amount
$t_2 - t_1$	12 hr − 1 hr	the denominator, 11 hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{A1 - A2}{t_2 - t_1}$$

For zero order the amount falls by the same quantity each hour, $A = A0 - k_0\, t$, so k_0, the zero-order rate constant, is the constant amount lost per unit time, the slope of amount against time: the difference between two amounts over the time between them, in mg/hr.

ALGEBRA

$$k_0 = \frac{478\ \text{mg} - 230\ \text{mg}}{12\ \text{hr} - 1\ \text{hr}} = \frac{248\ \text{mg}}{11\ \text{hr}} = 22.55\ \text{mg/hr}$$

478 mg at 1 hr minus 230 mg at 12 hr is 248 mg lost over 12 − 1 = 11 hr, the two points her key uses; $\frac{248\ \text{mg}}{11\ \text{hr}} = 22.55$ mg/hr. Milligrams over hours is an amount per unit time, the unit of a zero-order rate constant.

ROUND

$$k_0 = 22.5\ \text{mg/hr}$$

Her key gives 22.5 mg/hr, the 22.545 mg/hr rounded to one decimal place. The table is not perfectly straight: the 1 hr and 16 hr points give 22.53 mg/hr, the same to one decimal place.

DOES THE NUMBER MAKE SENSE?

From 3 hr (433 mg) to 6 hr (365 mg) the drop is about the same as from 9 to 12 hr, 298 to 230 mg, a nearly constant loss; 478 − 230 = 248 mg over 11 hr is 22.5 mg/hr.

Her homework with her own numbers; her key gives 22.5 mg/hr, the 22.545 mg/hr rounded to one decimal place.

2c Worked by the drill

Answer: 11.1 hr

WHAT THE STEM GIVES

A at 1 hr	478 mg	brought back 1 hr to A0 = 500.5 mg, rounded to 500 mg
table	478 mg at 1 hr and 230 mg at 12 hr	gave k_0 = 22.5 mg/hr in part b, in the denominator $2k_0$

THE WORKING, ONE LINE AT A TIME

SETUP

$$A0 = A + k_0\,t = 478\text{ mg} + (22.545\text{ mg/hr})(1\text{ hr}) = 500.5\text{ mg}$$

The zero-order line $A = A0 - k_0\,t$ rearranges to $A0 = A + k_0\,t$. Her key starts from the 1 hr entry, 478 mg, which has already lost (22.545 mg/hr)(1 hr) = 22.545 mg; adding it back gives A0 = 500.5 mg, which her key rounds to 500 mg.

ALGEBRA

$$t½ = \frac{A0}{2k_0} = \frac{500\text{ mg}}{2(22.5\text{ mg/hr})} = 11.11\text{ hr}$$

Half of A0 is 250 mg, and at 22.5 mg/hr that loss takes $\frac{250\text{ mg}}{22.5\text{ mg/hr}} = 11.11$ hr, which is $t½ = \frac{A0}{2k_0}$. Unlike first order, this half-life grows with the starting amount, because a constant loss takes longer to remove half of a larger amount.

ROUND

$$t½ = 11.1\text{ hr}$$

Her key gives 11.1 hours. The unrounded 500.5 mg and 22.545 mg/hr give $\frac{500.5\text{ mg}}{2(22.545\text{ mg/hr})} = 11.10$ hr, the same to one decimal place.

> **DOES THE NUMBER MAKE SENSE?**
> Half of A0 = 500 mg lies between the 298 mg at 9 hr and the 230 mg at 12 hr, so t½ is between 9 and 12 hr: 11.1 hr.

Her homework with her own numbers; her key gives 11.1 hours, from A0 rounded to 500 mg and k_0 to 22.5 mg/hr.

3a Worked by the drill

Answer: 14.5 days

WHAT THE STEM GIVES

C_0	315 mg/mL	the numerator of the ratio inside the logarithm
t	30 days	the denominator of k
temperature	25°C	*not needed*
C	75 mg/mL	the denominator of the ratio inside the logarithm

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(315/75)}{30\ \text{days}} = \frac{1.4351}{30\ \text{days}} = 0.04784\ \text{day}^{-1}$$

First order: $\ln C = \ln C_0 - kt$, so $k = \frac{\ln(C_0/C)}{t}$, the log of the ratio of starting to final concentration over the time between them. $\frac{315}{75} = 4.200$, a pure number because the mg/mL cancel, and $\ln 4.200 = 1.4351$; the time is in days, so k is in day^{-1}.

ALGEBRA

$$t½ = \frac{0.693}{0.04784\ \text{day}^{-1}} = 14.49\ \text{days}$$

$t½ = \frac{0.693}{k}$ because $k\ t½ = \ln 2$ when C falls to half; the reciprocal days in k invert to days: $\frac{0.693}{0.04784\ \text{day}^{-1}} = 14.49$ days. A first-order half-life does not depend on the starting concentration: every halving takes the same 14.49 days.

ROUND

$$t½ = 14.5\ \text{days}$$

Her key gives 14.5 days, the 14.49 days rounded to one decimal place. Under first order every halving takes this long, whatever the concentration it starts from.

> **DOES THE NUMBER MAKE SENSE?**
> 315 to 75 mg/mL is a fall to under a quarter in 30 days, more than two halvings, so t½ is under 15 days: 14.5 days.

Her homework with her own numbers; her key gives this answer.

3b Worked by the drill

Answer: 19.7 days

WHAT THE STEM GIVES

C_0	315 mg/mL	in k_0 and the numerator of $\frac{C_0}{2k_0}$
t	30 days	the denominator of k_0
temperature	25°C	*not needed*
C	75 mg/mL	subtracted from C_0 in k_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_0 = \frac{315 - 75\ \text{mg/mL}}{30\ \text{days}} = 8.0000\ \text{(mg/mL)/day}$$

Zero order: $C = C_0 - k_0 t$, so $k_0 = \frac{C_0 - C}{t}$, the constant loss per unit time. $315 - 75 = 240$ mg/mL was lost over 30 days, so $k_0 = 8.0000$ (mg/mL)/day, a concentration per unit time rather than the day^{-1} of a first-order k.

ALGEBRA

$$t_{1/2} = \frac{C_0}{2k_0} = \frac{315\ \text{mg/mL}}{2(8.0000\ (\text{mg/mL})/\text{day})} = 19.69\ \text{days}$$

Half of the original 315 mg/mL is 157.5 mg/mL; at a constant 8.0000 (mg/mL)/day that loss takes $\frac{157.5\ \text{mg/mL}}{8.0000\ (\text{mg/mL})/\text{day}} = 19.69$ days, which is $t_{1/2} = \frac{C_0}{2k_0}$. The mg/mL cancel and days are left; the zero-order half-life uses the starting concentration, so it differs from the first-order 14.49 days of part a.

ROUND

$t_{1/2} = 19.7$ days

Her key gives 19.7 days, the 19.69 days rounded to one decimal place. The same two points give 14.5 days under first order, so the order decides the half-life.

> **DOES THE NUMBER MAKE SENSE?**
>
> Zero order loses the same amount each day. In 30 days more than three-quarters of the 315 mg/mL was lost, so halving takes less than two-thirds of 30 days, under 20 days: 19.7 days.

Her homework with her own numbers; her key gives this answer.